CUSTOMER RETENTION & CHURN ANALYSIS

A Data Analytics Project Report
Internship Programme: Future Interns – Data Science & Analytics
Project: Task 2 – Customer Retention & Churn Analysis
Dataset: IBM Telco Customer Churn

Prepared as a structured analysis of customer churn patterns, retention indicators, and practical business recommendations.

Executive Summary

Customer churn is an important business concern because losing existing customers can reduce recurring revenue and increase the effort required to acquire replacement customers. This project analyses the IBM Telco Customer Churn dataset to identify customer groups associated with higher churn and to translate the findings into practical retention recommendations.
The dataset contains 7,043 customer records and 21 original columns. The analysis found that 1,869 customers had churned and 5,174 had remained, giving an overall churn rate of 26.54% and a retention rate of 73.46%. Contract type, payment method, internet service, tenure, and selected support services showed notable differences in churn rates.
The analysis was carried out using Python in Jupyter Notebook, with Pandas and NumPy for data handling and Matplotlib and Seaborn for visual exploration. A Power BI dashboard was developed to present the main churn metrics and customer-segment comparisons.

Important interpretation: the dataset is a historical snapshot. It does not provide explicit customer-stated churn reasons or a month-by-month event history. The results therefore describe associations in the available data, not proven causes of churn.

1. Introduction

Customer retention refers to an organisation's ability to maintain its relationship with existing customers over time. Churn occurs when a customer stops using a service or ends a subscription. Understanding which customer groups have higher churn rates can help an organisation prioritise retention activities and improve the customer experience.
This project uses a telecommunications customer dataset to examine how churn varies across contract arrangements, payment methods, service subscriptions, customer tenure, and demographic or household attributes. The objective is descriptive and diagnostic: to identify patterns that may help guide further investigation and retention planning.

2. Project Objectives

The main objectives are:

- Measure the overall number and proportion of churned and retained customers.
- Compare churn rates across contract types, payment methods, and internet service categories.
- Examine the relationship between tenure and churn status.
- Explore differences associated with support services and selected customer characteristics.
- Build a Power BI dashboard that communicates the findings clearly.
- Develop actionable recommendations while acknowledging the limits of the dataset.

3. Dataset Description

The project uses the IBM Telco Customer Churn dataset, obtained from Kaggle. It contains 7,043 rows and 21 columns. Each row represents an individual customer, with fields describing customer attributes, subscribed services, account information, charges, tenure, and churn status.

Dataset attribute	Description
Source	IBM Telco Customer Churn dataset (Kaggle)
Records	7,043 customer records
Original features	21 columns
Target variable	Churn (Yes / No)
Examples of fields	tenure, Contract, Payment Method, Internet Service, Monthly Charges, Total Charges, Online Security, Tech Support

The dataset is a static customer snapshot. Tenure is available as a customer-level duration measure, but the data does not include a sequence of monthly observations for each customer. Consequently, the project analyses tenure groups rather than claiming to show month-by-month retention trends.

4. Tools and Technologies

Tool / library	Role in the project
Python	Primary programming language for data analysis.
Jupyter Notebook	Environment used to execute and document the analysis.
Pandas and NumPy	Data loading, cleaning, transformation, grouping, and summary calculations.
Matplotlib and Seaborn	Charts and exploratory visualisations.
Power BI Desktop	Interactive dashboard and presentation of customer churn metrics.

5. Methodology

The project followed a structured analytics workflow, beginning with dataset inspection and cleaning, followed by exploratory analysis, customer-segment comparisons, dashboard development, and recommendations.

5.1. Data loading and inspection

The CSV dataset was loaded into a Pandas Data Frame. The initial inspection covered the dataset dimensions, column names, data types, missing values, and duplicate records. This established the data quality baseline before analysis.

5.2. Data cleaning and preprocessing

The Total Charges field contained 11 blank entries. These records had tenure equal to zero and were associated with customers who had not churned. The blank strings were converted to numeric values and filled with zero so that the column could be used in numeric analysis. The dataset was then checked for missing values and duplicate rows.

Quality check	Result
Initial dataset shape	7,043 rows × 21 columns
Blank Total Charges entries	11
Treatment	Converted Total Charges to numeric; blank values filled with 0 for zero-tenure records
Missing values after cleaning	0
Duplicate rows	0

Additional analysis fields were created for tenure grouping and estimated customer value. The tenure groups used were 0–12 months, 13–24 months, 25–48 months, and 49–72 months. The estimated customer value was calculated as Monthly Charges multiplied by tenure; this is a simple historical proxy and should not be interpreted as a predictive customer lifetime value model.

5.3. Exploratory data analysis

Churn was compared across categorical fields such as Contract, Payment Method, Internet Service, Online Security, and Tech Support. Numeric fields such as tenure and Monthly Charges were summarised by churn status. Customer characteristics such as Senior Citizen, Partner, and Dependents were also examined.

5.4. Dashboard development

A Power BI dashboard was developed using the cleaned customer dataset. It includes headline customer and churn metrics, a churn distribution visual, comparisons by contract and payment method, internet-service segmentation, tenure-group churn rates, a tenure-versus-monthly-charges scatter plot, contract distribution, and a matrix comparing churned customer counts across Senior Citizen and Partner categories.

6. Key Findings

6.1. Overall churn and retention

Metric	Value
Total customers	7,043
Churned customers	1,869
Retained customers	5,174
Churn rate	26.54%
Retention rate	73.46%

Approximately one in four customers in the dataset had churned. This overall rate provides the baseline against which customer-segment churn rates can be compared.

6.2. Contract type

Contract type	Churn rate	Retention rate
Month-to-month	42.71%	57.29%
One year	11.27%	88.73%
Two years	2.83%	97.17%

Month-to-month customers had the highest churn rate at 42.71%. Customers with one-year and two-year contracts had substantially lower churn rates. This pattern suggests that contract type is strongly associated with customer retention in this dataset. It does not, by itself, prove that changing a customer's contract will prevent churn.

6.3. Payment method

Payment method	Churn rate
Electronic check	45.29%
Mailed check	19.11%
Bank transfer (automatic)	16.71%
Credit card (automatic)	15.24%

Electronic check users had the highest churn rate at 45.29%. The three other payment categories had lower churn rates. The organisation could investigate whether payment friction, customer preferences, or differences in customer profiles are related to this pattern before selecting an intervention.

6.4. Internet service

Internet service	Churn rate
Fiber optic	41.89%
DSL	18.96%
No internet service	7.40%

Fiber optic customers had a higher churn rate than DSL customers and customers without internet service. This segment may warrant further investigation into service quality, pricing, customer expectations, and support experience. The dataset does not identify which of these factors, if any, explains the observed difference.

6.5. Tenure and churn

Churn status	Customer count	Mean tenure	Median tenure
Retained (No)	5,174	37.57 months	38 months
Churned (Yes)	1,869	17.98 months	10 months

Customers who churned had a considerably shorter average tenure than customers who remained. The median tenure of churned customers was 10 months, compared with 38 months for retained customers. This highlights the importance of examining the early customer relationship and onboarding experience.

6.6. Monthly charges

Churn status	Mean monthly charges	Median monthly charges
Retained (No)	$61.27	$64.425
Churned (Yes)	$74.44	$79.65

Customers who churned had higher average and median monthly charges than retained customers. This may indicate that higher monthly cost is associated with churn in the dataset. Additional analysis would be needed to separate price effects from differences in service bundles or customer profiles.

6.7. Customer characteristics

Customer segment	Churn rate
Senior citizens	41.68%
Non-senior customers	23.61%
Customers without a partner	32.96%
Customers with a partner	19.66%
Customers without dependents	31.28%
Customers with dependents	15.45%

The observed churn rate was higher among senior citizens, customers without a partner, and customers without dependents. These are descriptive segment differences and should be used carefully. They do not justify assumptions about an individual customer or imply that demographic attributes cause churn.

6.8. Online security and technical support

Service status	Churn rate
OnlineSecurity = No	41.77%
OnlineSecurity = Yes	14.61%
TechSupport = No	41.64%
TechSupport = Yes	15.17%

Customers without online security or technical support had higher churn rates than customers subscribed to these services. This suggests that support-service adoption is a useful segment for further retention analysis. The relationship may also reflect differences in customer needs, service bundles, or other factors.

7. Business Insights and Recommendations

Finding	Suggested action
Month-to-month churn is 42.71%.	Offer suitable longer-term plans, explain contract benefits clearly, and provide renewal incentives without pressuring customers.
Electronic check churn is 45.29%.	Investigate payment-related friction and communicate the convenience of automatic payment options.
Fiber optic churn is 41.89%.	Review service quality, installation experience, pricing, and complaint patterns for this segment.
Churned customers have shorter tenure.	Strengthen onboarding, early-life support, and proactive check-ins during the first year.
Higher monthly charges are associated with churn.	Review plan-value communication and identify customers who may benefit from suitable plan adjustments.
Customers without online security or technical support show higher churn.	Improve awareness of available support and security services and assess whether service gaps contribute to dissatisfaction.

Recommendations should be tested through controlled campaigns or further analysis before being treated as proven churn-reduction strategies. Customer feedback, complaints, service-quality records, and pricing history would help validate the proposed actions.

8. Power BI Dashboard Overview

The Power BI dashboard consolidates the analysis into a visual summary designed to support quick exploration. Its KPI cards present total customers, churned customers, retained customers, churn rate, retention rate, and average monthly charges.

The dashboard's supporting visuals include:

- Donut chart – distribution of churned and retained customers.
- Contract comparison – churned customer counts by contract type.
- Payment-method comparison – churned customer counts by payment method.
- 100% stacked column chart – churn composition across internet-service categories.
- Tenure-group line chart – churn rate across tenure groups.
- Scatter plot – customer tenure and monthly charges by churn status.
- Tree map – customer distribution by contract type.
- Matrix – churned customer counts by Senior Citizen and Partner categories.

The dashboard is intended as an exploratory reporting tool. Segment-level visuals should be interpreted alongside the underlying customer counts and the definitions of the measures use

9. Project Deliverables

Folder / file	Purpose
notebooks/Customer_Churn_Analysis.ipynb	Python notebook containing the data preparation, analysis, and visual exploration.
outputs/cleaned_customer_churn.csv	Cleaned dataset with derived analysis fields.
outputs/overall_churn_summary.csv	Overall customer and churn summary.
outputs/churn_factor_summary.csv	Summary of churn patterns across selected factors.
outputs/tenure_retention_summary.csv	Retention summary across tenure groups.
outputs/business_recommendations.csv	Business findings and recommended actions.
dashboard/Customer_Churn_Dashboard.pbix	Power BI dashboard file.
screenshots/	Screenshots of the dashboard and selected visuals.

10. Limitations and Considerations

- No explicit churn reasons: the dataset records churn status but does not provide a customer-stated reason for leaving.
- Static snapshot: the dataset does not contain monthly event records, so it cannot support a true month-by-month cohort or time-series churn analysis.
- Association is not causation: segment differences do not prove that a particular attribute causes churn.
- Limited context: service quality, customer complaints, competitor offers, and customer feedback are not included in the available fields.
- Estimated customer value: Monthly Charges × tenure is a simple historical estimate, not a validated predictive customer lifetime value measure.

11. Conclusion

This project examined customer churn in the IBM Telco Customer Churn dataset using Python-based exploratory data analysis and a Power BI dashboard. The overall churn rate was 26.54%. Higher churn rates were observed among month-to-month contract customers, electronic check users, and fibre optic customers. Churned customers also had shorter tenure and higher average monthly charges than retained customers. Differences were observed across selected customer characteristics and support-service subscriptions.

The findings provide a starting point for targeted retention analysis. The most practical next steps are to investigate early-tenure customer experience, review the needs of high-churn segments, and validate potential interventions using additional operational data or controlled experiments. The project demonstrates an end-to-end workflow covering data cleaning, exploratory analysis, dashboard reporting, and business recommendations.

12. Dataset Reference

IBM Telco Customer Churn dataset, available through Kaggle: https://www.kaggle.com/datasets/blastchar/telco-customer-churn

End of Report